



Alejandro Moscoso Deossa

Data Scientist · ML Engineer · AI Enthusiast

alejo-mdm@hotmail.com · (+57) 312 239 69 42 · linkedin.com/in/alejomd17 · Medellín, Colombia

PROFESSIONAL SUMMARY

Full-stack data professional skilled across the entire AI/ML lifecycle — from exploratory analysis and model development to data engineering orchestration, scalable cloud architecture, and building data platforms that serve advanced analytics and AI. Combines 8+ years of experience spanning financial risk, econometrics, NLP, forecasting, and cloud-native solutions, with deep expertise in Python, SQL, PySpark, and the Azure ecosystem (Databricks, Data Factory, Azure ML). Solid academic foundation including a Master's in Economics and Financial Engineering. Currently expanding into Generative AI and Agentic workflows: RAGs, LangChain, and Prompt Engineering.

EXPERIENCE

Data Scientist | Machine Learning Engineer | AI Enthusiast | Data Analyst | Statistical & Financial Analyst | Data Engineer

- **Data Science:** Designed and executed end-to-end data science projects — data acquisition, exploratory analysis, feature engineering, and statistical modeling — to generate actionable insights across financial, commercial, and operational domains.
- **Machine Learning:** Built and deployed predictive and prescriptive ML models using Scikit-learn, XGBoost, Prophet, Statsmodels TensorFlow and PyTorch covering regression, classification, clustering, and time-series forecasting. Implemented Reinforcement Learning for decision-making systems. Conducted A/B tests and quasi-experiments to measure model uplift and validate business hypotheses.
- **Deep Learning & NLP:** Developed Transformer-based NLP pipelines (Hugging Face) for sentiment analysis, demand forecasting, and text classification; applied Transfer Learning for domain-specific models; used semantic similarity and word embeddings to detect anomalies and identify patterns across massive datasets.
- **Data Engineering:** Designed scalable ETL/ELT pipelines and Data Lakehouse architectures using Azure Databricks (Delta Lake, PySpark, Apache Spark) and Azure Data Factory; managed high-volume data ingestion across relational and non-relational cloud databases; handled diverse data formats (JSON, CSV, XLSX) for model integration and retrieval pipelines.
- **Cloud Azure & MLOps:** Deployed and monitored production models in Azure ML and Databricks with MLflow and CI/CD practices; orchestrated workflows via Azure Functions and Logic Apps; managed Azure DevOps for infrastructure, versioning, and continuous delivery. Applied cluster performance optimization for large-scale PySpark workloads.
- **Backend & APIs:** Built scalable REST APIs and batch-serving patterns using FastAPI to expose ML models and LLMs in production; developed automated data acquisition pipelines via web scraping (Beautiful Soup) and external API integrations.
- **Analytics & Optimization:** Implemented Linear and Mixed Integer Programming for supply chain optimization, pricing strategy, and demand forecasting; delivered revenue management and elasticity estimation models across commercial and financial domains.
- **Data Analysis & Visualization:** Processed and analyzed large datasets from multiple sources; built advanced dashboards in Power BI, Plotly, Matplotlib, and Seaborn; conveyed complex trends and analytical findings to diverse technical and non-technical stakeholders.
- **Business Intelligence:** Delivered end-to-end BI solutions — KPI tool development, dashboard administration, and Power BI environments — aligned with business requirements and organizational strategy.
- **Generative AI & Agentic Systems:** Designed agentic architectures and multi-agent workflows using LangChain and Prompt Engineering; built early-stage RAG solutions with Vector Databases and embeddings to improve information retrieval from internal documentation; integrated GenAI pipelines into existing cloud workflows.
- **Financial Risk Modeling:** Performed liquidity risk, credit risk, and market risk analysis; built comparative and counterparty risk indicators against market and financial sector benchmarks using R, Python, and Excel.
- **Econometric Forecasting:** Developed econometric models to track macroeconomic variables and forecast credit portfolio behavior, default rates, and loan performance; applied scenario analysis and stress-testing methodologies.
- **Data Analysis & KPI Dashboards:** Designed and maintained Power BI dashboards with SQL data extraction to support organizational performance monitoring and executive reporting.
- **Budget & Cash Flow Management:** Led projection of receivables collection, supplier payments, and cash flow calculations; produced financial indicator analyses for receivables and cash management systems.
- **Cross-functional Support:** Collaborated with accounting, budgeting, accrual, invoicing, and foreign trade areas to deliver financial models, market evaluations, and institutional budget planning reports.

Companies: Summa SCI – Grupo Argos (May 2023 – Curr) · 2B eCommerce (Apr 2021 – May 2023) · Grupo Réditos (Dec 2019 – Apr 2021) · Comedal – Cooperativa (Ene 2017 – Nov 2019) · Plaza Mayor (Aug 2014 – Dec 2016)

TECHNICAL SKILLS & TOOLS

Programming & Databases

Python, SQL, R, PySpark, HTML/CSS
Microsoft SQL Server, Delta Lake, Data Lakehouse

Cloud & Data Engineering

Microsoft Azure, Azure Data Factory, Azure Databricks
Azure ML, Azure Functions, Logic Apps, MLOps
Azure DevOps, CI/CD, Apache Spark, PySpark

Machine Learning & Data Science

Scikit-learn, XGBoost, TensorFlow, PyTorch, PuLP, SciPy
NumPy, Pandas, Seaborn, Reinforcement Learning
Statistical modeling, econometrics, financial risk

Generative AI & NLP

LLMs, RAG, LangChain, Prompt Engineering
Hugging Face Transformers, embeddings, vector DBs

Visualization & Business Intelligence

Power BI, Plotly, Matplotlib, Seaborn
Advanced Excel (VBA), KPI dashboards, reporting

Development & DevOps

FastAPI, Git, GitHub, BeautifulSoup, web scraping
Databricks notebooks, MLflow, batch pipelines

Languages

Spanish (Native) · English (Advanced)

CERTIFICATIONS & ACCREDITATIONS

Databricks Lakehouse · Databricks · 2024
Databricks Fundamentals · Databricks · 2024
Azure Fundamentals · Platzi · 2023
Transfer Learning – Hugging Face · Platzi · 2023

NLP with Python & NLTK · Platzi · 2023
Practical Machine Learning · Platzi · 2021
Business Analysis from Data Science · Platzi · 2021
Project Risk Management · BID · 2019

EDUCATION

Master in Economics

Universidad de Antioquia
2019 – 2022

Major in Statistics & Financial Risk

Universidad de Medellín
2016 – 2017

Financial Engineering

Universidad de Medellín
2010 – 2015